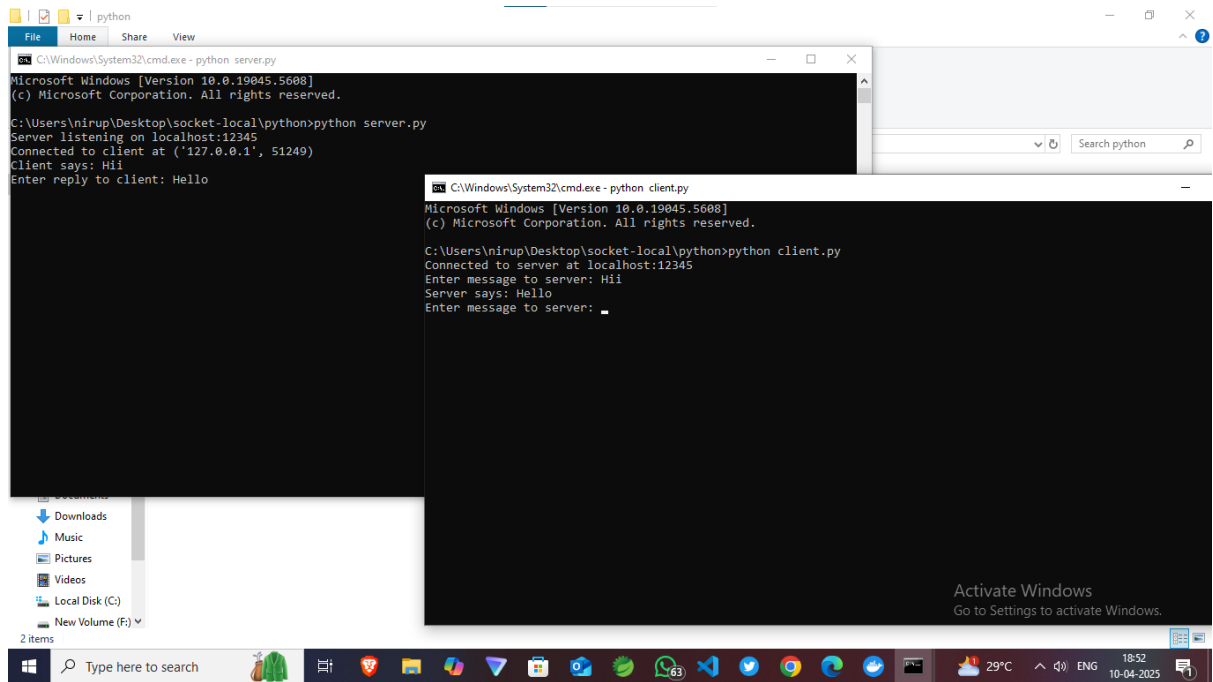


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Exercise - 1

➤ Python Local



The screenshot shows a Windows 10 desktop with two command prompts open. The left prompt is titled 'C:\Windows\System32\cmd.exe - python_server.py' and shows the execution of 'python server.py'. It displays 'Server listening on localhost:12345', 'Connected to client at ('127.0.0.1', 51249)', 'Client says: Hii', and 'Enter reply to client: Hello'. The right prompt is titled 'C:\Windows\System32\cmd.exe - python_client.py' and shows the execution of 'python client.py'. It displays 'Connected to server at localhost:12345', 'Enter message to server: Hii', 'Server says: Hello', and 'Enter message to server:'. The taskbar at the bottom shows various application icons and the system clock indicating 18:52 on 10-04-2025.

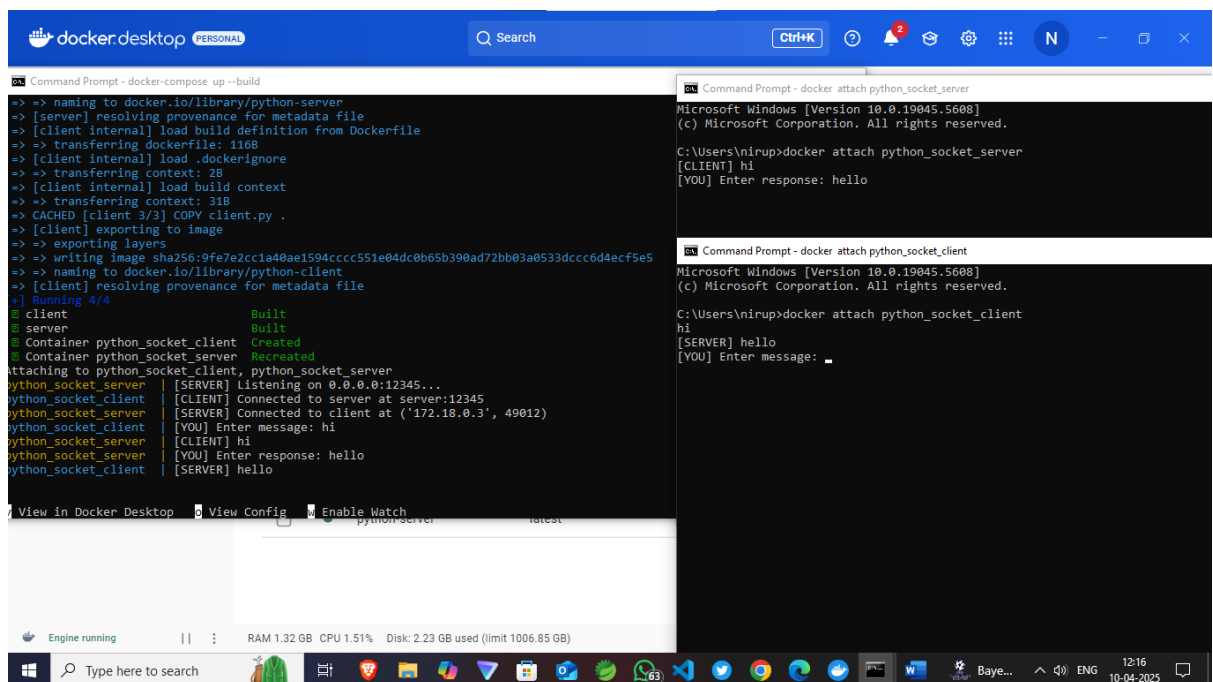
```
C:\Windows\System32\cmd.exe - python_server.py
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. All rights reserved.

C:\Users\nirup\Desktop\socket-local\python>python server.py
Server listening on localhost:12345
Connected to client at ('127.0.0.1', 51249)
Client says: Hii
Enter reply to client: Hello

C:\Windows\System32\cmd.exe - python_client.py
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. All rights reserved.

C:\Users\nirup\Desktop\socket-local\python>python client.py
Connected to server at localhost:12345
Enter message to server: Hii
Server says: Hello
Enter message to server: 
```

➤ Python Docker



The screenshot shows the Docker Desktop interface with three command prompts. The first prompt, titled 'Command Prompt - docker-compose up --build', shows the build process for a Docker service named 'python-socket'. It lists the steps: naming to docker.io/library/python-server, resolving provenance for metadata file, transferring dockerfile: 116B, loading .dockerignore, transferring context: 2B, loading build context, transferring context: 31B, copying client.py, exporting to image, and writing image sha256:9fe7e2cc1a40ae1594cccc551e04dc0b65b390ad72bb03a0533dccc6d4ecf5e5. The second prompt, titled 'Command Prompt - docker attach python_socket_server', shows the output of 'docker attach python_socket_server', displaying '[SERVER] listening on 0.0.0.0:12345...', '[CLIENT] Connected to server at server:12345', '[SERVER] Connected to client at ('172.18.0.3', 49012)', '[YOU] Enter message: hi', '[CLIENT] hi', '[YOU] Enter response: hello', and '[SERVER] hello'. The third prompt, titled 'Command Prompt - docker attach python_socket_client', shows the output of 'docker attach python_socket_client', displaying '[SERVER] hello' and '[YOU] Enter message:'. The Docker Desktop interface also shows a list of containers and the system status at the bottom.

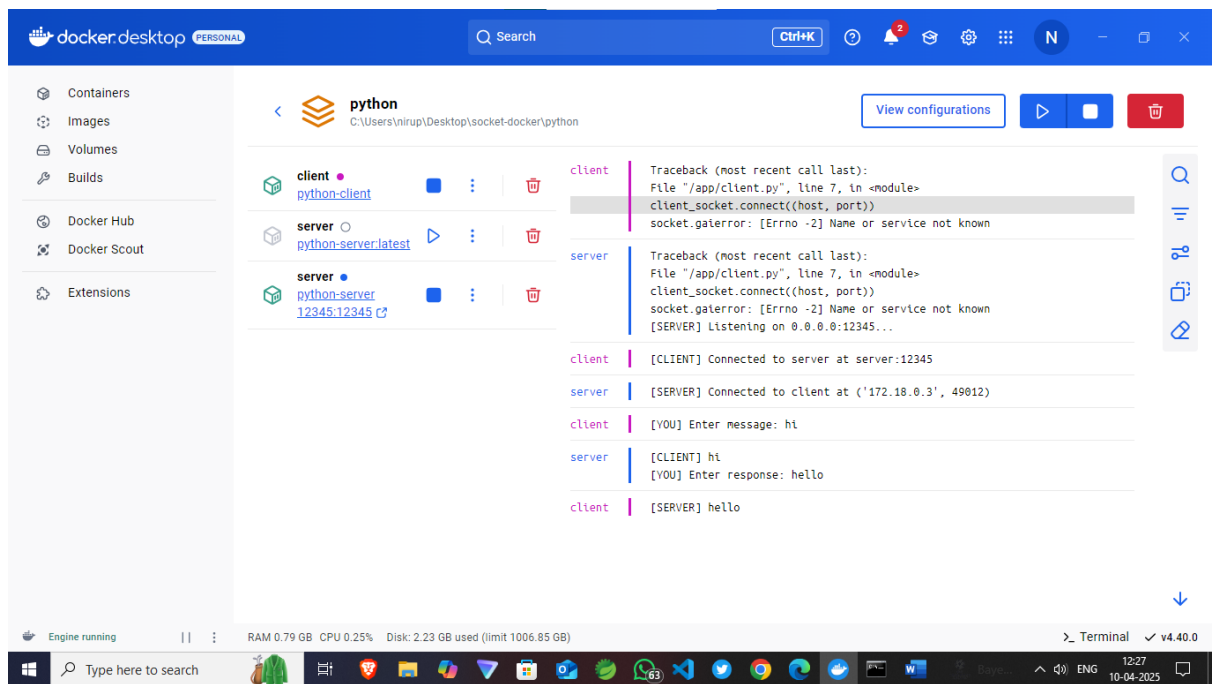
```
Command Prompt - docker-compose up --build
-> => naming to docker.io/library/python-server
-> [server] resolving provenance for metadata file
-> [client internal] load build definition from Dockerfile
-> => transferring dockerfile: 116B
-> [client internal] load .dockerignore
-> => transferring context: 2B
-> [client internal] load build context
-> => transferring context: 31B
-> CACHED [client 3/3] COPY client.py .
-> [client] exporting to image
-> => exporting layers
-> => writing image sha256:9fe7e2cc1a40ae1594cccc551e04dc0b65b390ad72bb03a0533dccc6d4ecf5e5
-> => naming to docker.io/library/python-client
-> [client] resolving provenance for metadata file
4) Running 4/4
client Built
server Built
Container python_socket_client Created
Container python_socket_server Recreated
Attaching to python_socket_client, python_socket_server
python_socket_server | [SERVER] listening on 0.0.0.0:12345...
python_socket_client | [CLIENT] Connected to server at server:12345
python_socket_server | [SERVER] Connected to client at ('172.18.0.3', 49012)
python_socket_client | [YOU] Enter message: hi
python_socket_server | [CLIENT] hi
python_socket_server | [YOU] Enter response: hello
python_socket_client | [SERVER] hello

Command Prompt - docker attach python_socket_server
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. All rights reserved.

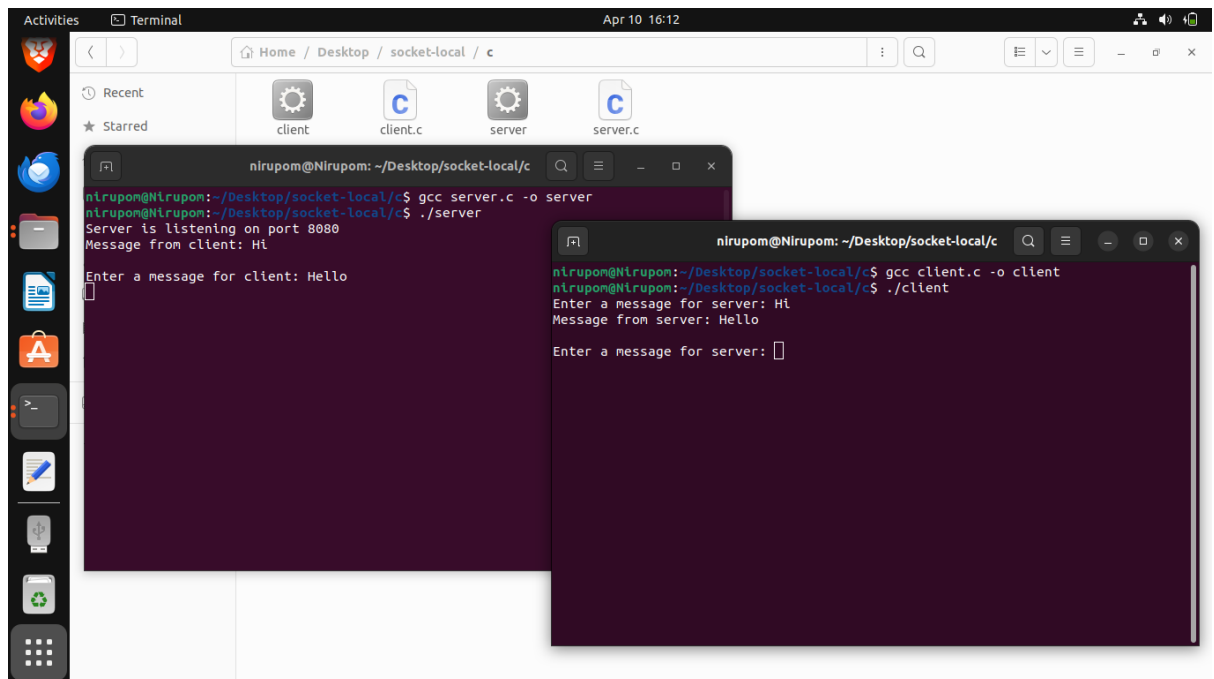
C:\Users\nirup>docker attach python_socket_server
[CLIENT] hi
[YOU] Enter response: hello

Command Prompt - docker attach python_socket_client
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. All rights reserved.

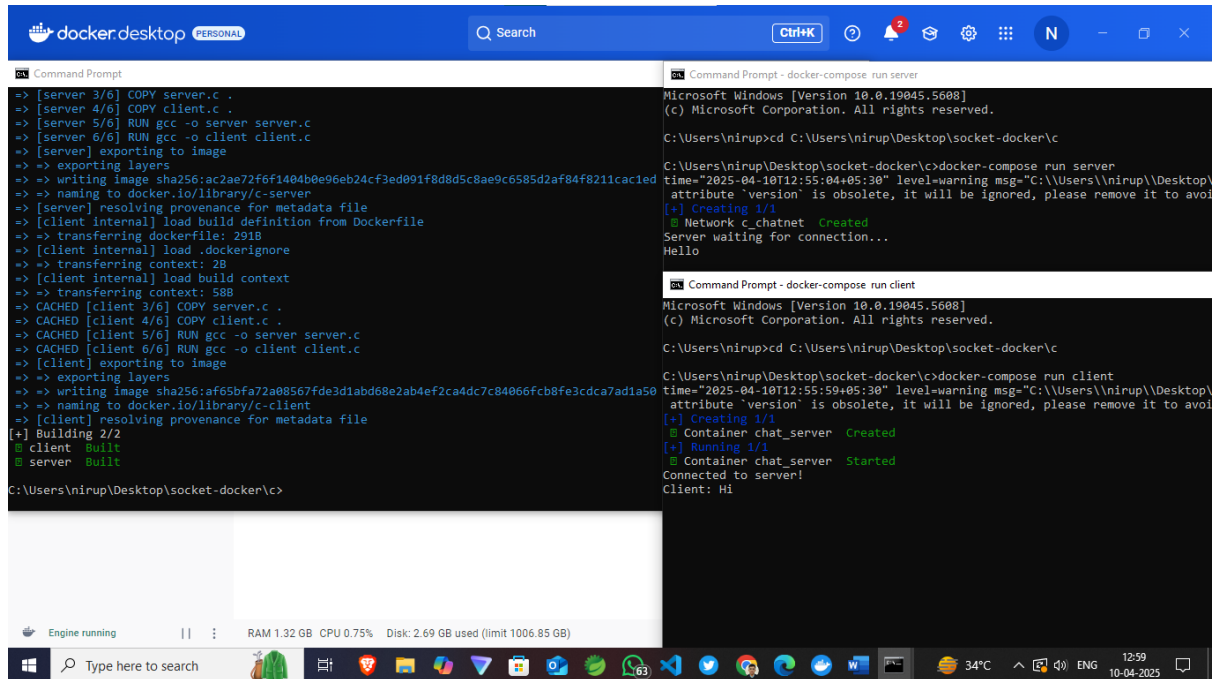
C:\Users\nirup>docker attach python_socket_client
hi
[SERVER] hello
[YOU] Enter message: 
```



➤ C Local



➤ C Docker



The screenshot shows the Docker Desktop interface with two terminal windows open. The left terminal shows the process of building a Docker image for a C server and client. The right terminal shows the execution of the Docker Compose command to run the containers.

```
[+] Building 2/2
 client Built
 server Built

C:\Users\nirup\Desktop\socket-docker>
```

```
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. All rights reserved.

C:\Users\nirup>cd C:\Users\nirup\Desktop\socket-docker\c

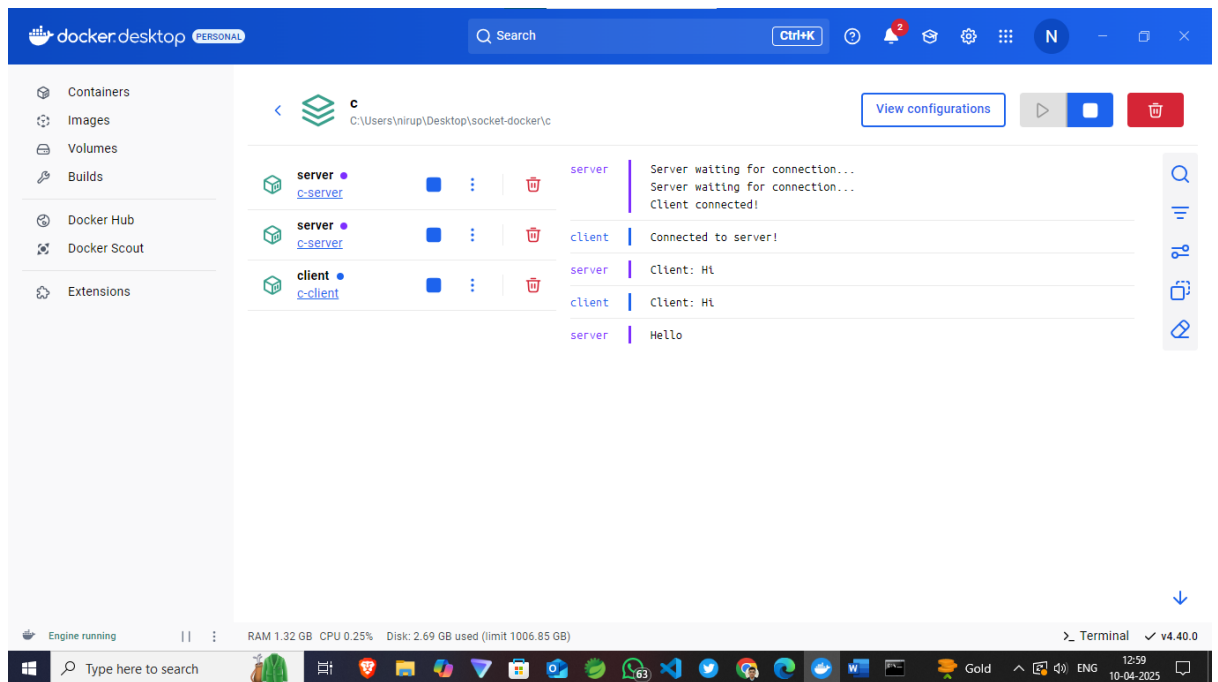
C:\Users\nirup\Desktop\socket-docker\c>docker-compose run server
time="2025-04-10T12:55:04+05:30" level=warning msg="C:\\Users\\nirup\\Desktop\\
 attribute 'version' is obsolete, it will be ignored, please remove it to avoid
[+] Creating 1/1
  Network c_chatnet Created
Server waiting for connection...
Hello
```

```
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. All rights reserved.

C:\Users\nirup>cd C:\Users\nirup\Desktop\socket-docker\c

C:\Users\nirup\Desktop\socket-docker\c>docker-compose run client
time="2025-04-10T12:55:59+05:30" level=warning msg="C:\\Users\\nirup\\Desktop\\
 attribute 'version' is obsolete, it will be ignored, please remove it to avoid
[+] Creating 1/1
  Container chat_server Created
[+] Running 1/1
  Container chat_server Started
Connected to server!
Client: Hi
```

Engine running | RAM 1.32 GB CPU 0.75% Disk: 2.69 GB used (limit 1006.85 GB)



The screenshot shows the Docker Desktop interface with the container management view. The left sidebar shows the navigation menu. The main area displays the container list and the logs for the selected container.

Containers

Images

Volumes

Builds

Docker Hub

Docker Scout

Extensions

View configurations

Container Name	Status	Logs
server	Running	Server waiting for connection... Server waiting for connection... Client connected!
client	Running	Connected to server!
server	Running	Client: Hi
client	Running	Client: Hi
server	Running	Hello

Engine running | RAM 1.32 GB CPU 0.25% Disk: 2.69 GB used (limit 1006.85 GB)

Terminal v4.40.0